SUMS Proofs Workshop: Fall Solutions

Note: Problems marked with (*) are difficult. (**) indicates *very difficult*.

Starter Problems

PROBLEM 1 Prove that 0 divides an integer a if and only if $a = 0$.

Proof. ($\Rightarrow$) If 0 divides a, there exists $k \in \mathbb{Z}$ with $0 \cdot k = a$. Hence $a = 0$.
($\Leftarrow$) If $a = 0$, then $0 = 0 \cdot k$ for any $k \in \mathbb{Z}$, so 0 divides a. $\square$

PROBLEM 2 Prove that there do not exist integers m and n such that $14m + 21n = 100$.

Proof. Suppose integers m, n satisfy $14m + 21n = 100$. Factor 7: $7(2m + 3n) = 100$, so $2m + 3n = 100/7$, which is not an integer. This is a contradiction. $\square$

PROBLEM 3 Prove that n^2 odd implies that n is odd.

Proof. We prove the contrapositive: if n is even then n^2 is even. If $n = 2k$ for some $k \in \mathbb{Z}$ then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$ is even. So if n^2 is odd, n cannot be even, so n is odd. $\square$

PROBLEM 4 Show that $\sqrt{2}$ is irrational.

Proof. Suppose $\sqrt{2} \in \mathbb{Q}$ so $\sqrt{2} = \frac{a}{b}$ in lowest terms with $a, b \in \mathbb{Z}$. Squaring gives $2b^2 = a^2$, so a^2 is even and hence a is even: $a = 2k$. Then $2b^2 = 4k^2$, so $b^2 = 2k^2$ and b is even. Thus both a and b are even, contradicting the assumption that the fraction was reduced and in lowest terms. Therefore $\sqrt{2}$ is irrational. $\square$

PROBLEM 5 Show that $1^2 + 2^2 + 3^2 + \cdots + n^2 = \dfrac{n(n+1)(2n+1)}{6}$ for every positive integer n.

Proof. Base case $n = 1$: $1^2 = 1 = \frac{1 \cdot 2 \cdot 3}{6}$. Assume true for $n - 1$. Then

$$\sum_{k=1}^{n} k^2 = \frac{(n-1)n(2n-1)}{6} + n^2 = \frac{2n^3 + 3n^2 + n}{6} = \frac{n(n+1)(2n+1)}{6},$$

completing the induction. $\square$

PROBLEM 6 For all $n \in \mathbb{N}$ show that $2^n \leq 3^n - 1$.

Proof. Base case $n = 1$: $2^1 = 2 \leq 3^1 - 1 = 2$. Assume $2^{n-1} \leq 3^{n-1} - 1$. Then

$$2^n = 2 \cdot 2^{n-1} \leq 2(3^{n-1} - 1) = 2 \cdot 3^{n-1} - 2.$$

Since $2 \cdot 3^{n-1} - 2 \leq 3^n - 1$ (because $3^n - 1 - (2 \cdot 3^{n-1} - 2) = 3^{n-1} + 1 > 0$), the inequality holds for n. Thus by induction $2^n \leq 3^n - 1$ for all $n \in \mathbb{N}$. $\square$

PROBLEM 7 Let $n \in \mathbb{Z}$. Suppose there exists $a \in \mathbb{Z}$ such that $n = a^2 - a$. Show that n is even.

Proof. Case 1: a even, $a = 2k$. Then $n = 4k^2 - 2k = 2(2k^2 - k)$ is even.
Case 2: a odd, $a = 2m + 1$. Then

$$n = (2m + 1)^2 - (2m + 1) = 4m^2 + 4m + 1 - 2m - 1 = 2(2m^2 + m),$$

which is even. In either case n is even. $\square$

PROBLEM 8 Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 3x^2 + 2$. Show that this is not injective.

Proof. Observe $f(1) = 3(1)^2 + 2 = 5$ and $f(-1) = 3(-1)^2 + 2 = 5$, so $f(1) = f(-1)$ with $1 \neq -1$. Hence f is not injective. $\square$

PROBLEM 9 Let $\sim$ be an equivalence relation on a set X. Prove that for all $a, b \in X$, $a \sim b$ if and only if they have equal equivalence classes.

Proof. ($\Rightarrow$) If $a \sim b$ and $x \in [a]$, then $x \sim a$ and by transitivity $x \sim b$, so $x \in [b]$. Thus $[a] \subseteq [b]$. Symmetrically $[b] \subseteq [a]$, so $[a] = [b]$.
($\Leftarrow$) If $[a] = [b]$ then $a \in [a] = [b]$, so $a \sim b$. $\square$

PROBLEM 10 Let $\sim$ be an equivalence relation on a set X. Prove that for all $a, b \in X$,

$$[a]_\sim = [b]_\sim \text{ or } [a]_\sim \cap [b]_\sim = \varnothing.$$

You may use the fact $[a]_\sim = [b]_\sim$ iff $a \sim b$ without proof (this is the previous problem!).

Proof. Let $a, b \in X$. There are two cases to consider.
Case 1: $a \sim b$. Then by the given fact, $[a]_\sim = [b]_\sim$.
Case 2: $a \not\sim b$. Suppose for contradiction $[a]_\sim \cap [b]_\sim \neq \varnothing$. Then $\exists c \in [a]_\sim \cap [b]_\sim$. This means $c \in [a]_\sim$ and $c \in [b]_\sim$, which implies $a \sim c$ and $b \sim c$. By symmetry of $\sim$, we have $c \sim b$. By transitivity of $\sim$, we get $a \sim b$, contradicting our assumption.
Therefore, either $[a]_\sim = [b]_\sim$ or $[a]_\sim \cap [b]_\sim = \varnothing$. $\square$

PROBLEM 11 Prove that

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots n \cdot n! = (n + 1)! - 1.$$

Note $n! = n \cdot (n - 1) \cdot (n - 2) \cdot 2 \cdot 1$.

Proof. We proceed by induction on n.
Base case: When $n = 1$, we have $1 \cdot 1! = 1 \cdot 1 = 1$ and $(1 + 1)! - 1 = 2! - 1 = 2 - 1 = 1$.
Inductive step: Assume the equation holds for $k \geq 1$, so

$$1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! = (k + 1)! - 1.$$

holds. We need to show it holds for $n = k + 1$. Expanding with the induction hypothesis,

$$\begin{aligned}
1 \cdot 1! + 2 \cdot 2! + \cdots + k \cdot k! + (k + 1) \cdot (k + 1)! &= ((k + 1)! - 1) + (k + 1) \cdot (k + 1)! \\
&= (k + 1)! - 1 + (k + 2 - 1)(k + 1)! \\
&= (k + 1)! - 1 + (k + 2)(k + 1)! - (k + 1)! \\
&= -1 + (k + 2)! \\
&= (k + 2)! - 1
\end{aligned}$$

Thus, the equation holds for $n = k + 1$, completing the proof by induction. $\square$

Problem 12 Define a relation $\sim$ on $\mathbb{N} \times \mathbb{N}$, where $0 \notin \mathbb{N}$, and

$$(a, b) \sim (c, d) \iff ad = bc.$$

Prove that $\sim$ is an equivalence relation.

Proof. We need to show that $\sim$ is reflexive, symmetric, and transitive.
Reflexivity: For any $(a, b) \in \mathbb{N} \times \mathbb{N}$, $ab = ba$ by commutativity, so $(a, b) \sim (a, b)$.
Symmetry: If $(a, b) \sim (c, d)$, then $ad = bc$. By commutativity, $bc = cb$ and $ad = da$, so $cb = da$, which means $(c, d) \sim (a, b)$.
Transitivity: Suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then $ad = bc$ and $cf = de$. Multiplying the first equation by f, $adf = bcf$. Multiplying the second equation by b, $bcf = bde$. Therefore, $adf = bde$. Since $d \neq 0$, dividing by d gives $af = be$, so $(a, b) \sim (e, f)$. □

Problem 13 Prove that if $|a + b| > 2$, then $|a| > 1$ or $|b| > 1$, where $a, b \in \mathbb{R}$.

Proof. We prove by contrapositive. Suppose $|a| \leq 1$ and $|b| \leq 1$. By the triangle inequality,

$$|a + b| \leq |a| + |b| \leq 1 + 1 = 2.$$

Since this is the contrapositive, we are done. □

Problem 14 Your friend claims that

$$1 + 2 + \cdots + n = \frac{n(n + 3)}{2} - 1.$$

The base case $n = 1$ holds. Does the claim hold in general? If so prove it, if not disprove it.

Proof. For $n = 1$, ${}^{1(1+3)}/_{2} - 1 = 1$, so the base case holds. For $n = 2$, the left side is $1 + 2 = 3$, while the right side is ${}^{2(2+3)}/_{2} - 1 = {}^{2\cdot 5}/_{2} - 1 = 5 - 1 = 4$. Since $3 \neq 4$, the claim is false. □

Problem 15 Suppose $a, b, c \in \mathbb{Z}$. Prove $a^2 + b^2 = c^2$ implies either a or b is even.

Proof. Suppose for contradiction that $a, b, c \in \mathbb{Z}$ such that $a^2 + b^2 = c^2$ and a, b are odd. So $\exists x, y \in \mathbb{Z}$ such that $a = 2x + 1$ and $b = 2y + 1$. Substituting:

$$\begin{aligned} c^2 &= (2x + 1)^2 + (2y + 1)^2 \\ &= 2(2(x^2 + y^2) + 2(x + y) + 1) \end{aligned}$$

We reach a contradiction, since $(2(x^2 + y^2) + 2(x + y) + 1)$ is odd and thus is not divisible by 2. Then $\sqrt{c^2}$ cannot be an integer. □

Problem 16 Suppose k, n are natural numbers, excluding 0. Then $k \nmid n$ if and only if there exist $q, r \in \mathbb{Z}$ with $0 < r < |k|$ such that $n = qk + r$. Hint: Use the division theorem: For any $n, k \in \mathbb{N}$, there exists unique $q, r \in \mathbb{N}$ such that $n = qk + r$ and $0 \leq r < |k|$.

Proof. By the division theorem, we know for any $n, k \in \mathbb{N}$, there exists unique $q, r \in \mathbb{N}$ such that $n = qk + r$ and $0 \leq r < |k|$. By the definition of being divisible, $k|n$ if and only if $\exists x \in \mathbb{Z}$ such that $n = xk$. Therefore, by the division theorem, $k|n$ if and only if $r = 0$. □

Problem 17 If $n, m \in \mathbb{N}$, prove that if $m|n$ and $n|m$ then $n = m$.

Proof. Since $m|n$, $\exists k \in \mathbb{Z}$ such that $km = n$. Similarly $\exists l \in \mathbb{Z}$ such that $ln = m$. Then $n = km = kln$. But then $kl = 1$. Since m, n can only be positive numbers, $k = 1$ and $l = 1$ so $m = n$. $\square$

Problem 18 For sets A, B, C show that $A \cap (B - C) = (A \cap B) - (A \cap C)$. Is it always true thar $A \cup (B - C) = (A \cup B) - (A \cup C)$?

Proof. Let $x \in A \cap (B \setminus C)$. Then $x \in A$, $x \in B$, and $x \notin C$. Thus $x \in A \cap B$ and, since $x \notin C$, also $x \notin A \cap C$. Hence $x \in (A \cap B) \setminus (A \cap C)$.
Conversely, let $x \in (A \cap B) \setminus (A \cap C)$. Then $x \in A$, $x \in B$, and $x \notin A \cap C$. Since $x \in A$, the latter means $x \notin C$. Thus $x \in A \cap (B \setminus C)$.
For the second part, Let $A = B = C = \{1\}$. Then

$$A \cup (B \setminus C) = \{1\}, \qquad (B \setminus C) = \varnothing, \qquad (A \cup B) \setminus (A \cup C) = \varnothing.$$

Thus the equality fails. $\square$

Problem 19 (From Berkeley's Intro to Proofs) Prove that the composition of any two decreasing functions is increasing. Hint: Given decreasing functions f and g, and numbers $x_1 < x_2$, show that $f(g(x_1)) < f(g(x_2))$.

Proof. Let $x_1 < x_2$. Since g is strictly decreasing we have $g(x_1) > g(x_2)$. Because f is strictly decreasing, from $g(x_1) > g(x_2)$ it follows that $f(g(x_1)) < f(g(x_2))$. Thus $x_1 < x_2 \Rightarrow f(g(x_1)) < f(g(x_2))$, so $f \circ g$ is strictly increasing. $\square$

Problem 20 Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$.

Proof. There are two cases.
Case 1: $x \leq y$. Then $\max(x, y) = y$ and $\min(x, y) = x$, so

$$\max(x, y) + \min(x, y) = y + x = x + y.$$

Case 2: $y < x$. Then $\max(x, y) = x$ and $\min(x, y) = y$, so

$$\max(x, y) + \min(x, y) = x + y.$$

In both cases the sum equals $x + y$, establishing the identity. $\square$

Harder Problems

Problem 21 (*) Assume that the following statement is true: if Micah dislikes pineapple pizza then he likes pineapple pizza. Does he like pineapple pizza?

Proof. Let

$$p: \text{ Micah dislikes pineapple pizza}$$

The hypothesis is $p \Rightarrow \neg p$. Then we can construct the following truth table:

p	$\neg p$	$p \implies \neg p$
T	T	T
F	F	T
F	T	T

Constructing the truth table shows the only way $p \Rightarrow \neg p$ can hold is when p is false (otherwise p and $\neg p$ would both be true or both false, impossible). Thus p is false, so $\neg p$ is true, i.e. Micah likes pineapple pizza. □

PROBLEM 22 (*) For functions $f : X \to Y$ and $g : Y \to X$ with $g \circ f = \mathrm{Id}_X$, decide the truth of:

1. f is surjective.
2. f is injective.
3. g is surjective.
4. g is injective.

Proof. (1) False. Counterexample: $X = [0, 1]$, $Y = [-1, 1]$, $f(x) = x$, $g(y) = |y|$. Then $g \circ f = \mathrm{Id}_{[0,1]}$ but f is not onto $[-1, 1]$.
(2) True. If $f(a) = f(b)$ then $a = g(f(a)) = g(f(b)) = b$, so f is injective.
(3) True. For any $x \in X$, let $y = f(x) \in Y$. Then $g(y) = g(f(x)) = x$, so g is surjective.
(4) False. Using the example in (1), $g(-1) = g(1) = 1$, so g need not be injective. □

PROBLEM 23 (*) Show that $\mathbb{N}$ has the same size as $\mathbb{Z}^+$ (the positive integers).

Proof. Define $f : \mathbb{N} \to \mathbb{Z}^+$ by $f(x) = x - 1$. Its inverse $g(y) = y + 1$ satisfies $g(f(x)) = x$ and $f(g(y)) = y$, so f is a bijection and the sets have the same cardinality. □

PROBLEM 24 (*) Negate the following definitions without using the word "not" or the symbol $\neg$:

1. $\forall x \in \mathbb{R} \setminus \mathbb{Q}$, if $x^2 \in S$ then $x^3 \in T$.
2. For all $x, y \in \mathbb{R}$, if $x, y \in \mathbb{Q}$ then $x + y \in \mathbb{Q}$.
3. We say $f : X \to Y$ is continuous at $p \in X$ if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. $\forall x \in E$ with $d_X(x, p) < \delta$, then $d_Y(f(x), f(p)) < \epsilon$ (†).
4. For some $f : X \to Y$, we say $\lim_{x \to p} f(x) = L$ if $\forall \epsilon$, $\exists \delta > 0$ s.t. $\forall x \in X$ with $d_X(x, p) < \delta$, then $d_Y(f(x), L) < \epsilon$ (†).

Proof. We write out the negations.

1. $\exists x \in \mathbb{R} \setminus \mathbb{Q}$ such that $x^2 \in S$ and $x^3 \notin T$.
2. $\exists x, y \in \mathbb{R}$ such that $x, y \in \mathbb{Q}$ and $x + y \notin \mathbb{Q}$.
3. We say $f : X \to Y$ is not continuous at $p \in X$ if $\exists \epsilon > 0$ such that $\forall \delta > 0$, $\exists x \in E$ with $d_X(x, p) < \delta$ and $d_Y(f(x), f(p)) \geq \epsilon$.

4. For some $f : X \to Y$, we say $\lim_{x \to p} f(x) \neq L$ if $\exists \epsilon > 0$ such that $\forall \delta > 0$, $\exists x \in X$ with $d_X(x, p) < \delta$ and $d_Y(f(x), L) \geq \epsilon$.

Negations can convoluted, so it's important to work through them methodically. $\square$

PROBLEM 25 (*) Consider the claim "$\mathcal{P}(A \cup B) \subseteq \mathcal{P}(A) \cup \mathcal{P}(B)$." If true, prove it. Otherwise, disprove it. Note that $\mathcal{P}(S)$ indicates the powerset of S.

Proof. The claim is false. We disprove by counterexample. Let $A = \{1\}$, $B = \{2\}$, and $A \cup B = \{1, 2\}$. Then $\{1, 2\} \in \mathcal{P}(A \cup B)$, but $\{1, 2\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$ because $\{1, 2\} \notin \mathcal{P}(A)$ and $\{1, 2\} \notin \mathcal{P}(B)$ since $\{1\} \notin \mathcal{P}(B)$ and $\{2\} \notin \mathcal{P}(A)$ respectively. $\square$

PROBLEM 26 (*) Define a sequence of numbers by $a_0 = 1$ and $a_{n+1} = 3a_n + 2$. Prove $a_n = 2 \cdot 3^n + 1$.

Proof. We use induction on n.
Base case: $a_0 = 2 \cdot 3^0 - 1 = 2 - 1 = 1$
Inductive step: Assume $a_k = 2 \cdot 3^k - 1$ for some $k \geq 0$. Then:

$$\begin{aligned} a_{k+1} &= 3a_k + 2 \\ &= 3(2 \cdot 3^k - 1) + 2 \\ &= 2 \cdot 3^{k+1} - 3 + 2 \\ &= 2 \cdot 3^{k+1} - 1 \end{aligned}$$

Therefore, $a_n = 2 \cdot 3^n - 1$ for all $n \geq 0$. $\square$

PROBLEM 27 (*) For sets A, B, C, is $(A \setminus B) \cup (B \setminus C) \subseteq A \setminus C$? If true prove it, else disprove it.

Proof. The claim is false. We provide a counterexample. Let $A = \{1, 2\}$, $B = \{2, 3\}$, and $C = \{2, 4\}$. Then $A \setminus B = \{1\}$ and $B \setminus C = \{3\}$, so $(A \setminus B) \cup (B \setminus C) = \{1, 3\}$. But $A \setminus C = \{1\}$, so $(A \setminus B) \cup (B \setminus C) \nsubseteq A \setminus C$. $\square$

PROBLEM 28 (*) Prove or give a counterexample: $(X \cap Y) \times (Z \cap W) = (X \times Z) \cap (Y \times W)$, for any sets X, Y, Z, W.

Proof. Let $(a, b) \in (X \cap Y) \times (Z \cap W)$. Then $a \in X \cap Y$ and $b \in Z \cap W$, which means $a \in X$, $a \in Y$, $b \in Z$, and $b \in W$. Thus, $(a, b) \in X \times Z$ and $(a, b) \in Y \times W$, which implies $(a, b) \in (X \times Z) \cap (Y \times W)$.
Conversely, let $(a, b) \in (X \times Z) \cap (Y \times W)$. Then $(a, b) \in X \times Z$ and $(a, b) \in Y \times W$, which means $a \in X$, $b \in Z$, $a \in Y$, and $b \in W$. Thus, $a \in X \cap Y$ and $b \in Z \cap W$, which implies $(a, b) \in (X \cap Y) \times (Z \cap W)$.
Therefore, $(X \cap Y) \times (Z \cap W) = (X \times Z) \cap (Y \times W)$ for any sets X, Y, Z, W. $\square$

PROBLEM 29 (*) Show that $a \sim b$ with $a, b, \in \mathbb{N} \iff$ the sum of the decimal digits of a and b are congruent mod 9 is an equivalence relation.

Proof. Let $D(n)$ be the sum of decimal digits of n. We need to verify reflexivity, symmetry, and transitivity of $\sim$.
Reflexivity: For any $a \in \mathbb{N}$, $D(a) \equiv D(a) \pmod 9$, so $a \sim a$.
Symmetry: If $a \sim b$, $D(a) \equiv D(b) \pmod 9$. This implies $D(b) \equiv D(a) \pmod 9$, so $b \sim a$.
Transitivity: If $a \sim b$ and $b \sim c$, then $D(a) \equiv D(b) \pmod 9$ and $D(b) \equiv D(c) \pmod 9$. By transitivity of congruence, $D(a) \equiv D(c) \pmod 9$, thus $a \sim c$.
Therefore, $\sim$ is an equivalence relation. □

PROBLEM 30 (*) Let $S = \{0,1\}^n$ be the set of n-digit binary strings. Define $h(x) =$ the number of 1s present in x, for $x \in S$ to be the *hamming weight* of x. Prove that $a \sim b \iff h(a) = h(b)$, for $a, b \in S$, is an equivalence relation.

Proof. We verify the three properties of an equivalence relation.
Reflexivity: For any $a \in S$, $h(a) = h(a)$, so $a \sim a$.
Symmetry: If $a \sim b$, $h(a) = h(b)$. By symmetry of equality, $h(b) = h(a)$, so $b \sim a$.
Transitivity: If $a \sim b$ and $b \sim c$, then $h(a) = h(b)$ and $h(b) = h(c)$. By transitivity of equality, $h(a) = h(c)$, thus $a \sim c$.
Therefore, $\sim$ is an equivalence relation on S. □

Challenging Problems

PROBLEM 31 (**) Show that

$$\bigcap_{n=1}^{\infty} \left(\frac{-1}{n}, \frac{1}{n} \right) = \{0\}.$$

Note: You may use without proof that, given any real number, there is another real number greater than it.

Proof. For any $n \geq 1$, we have $\frac{-1}{n} < 0 < \frac{1}{n}$, so $0 \in \left(\frac{-1}{n}, \frac{1}{n}\right)$ for all $n \geq 1$. Thus, $0 \in \bigcap_{n=1}^{\infty} \left(\frac{-1}{n}, \frac{1}{n}\right)$.
Let $x \neq 0$ be a real number. Then $|x| > 0$. Choose $N > \frac{1}{|x|}$. Then $\frac{1}{N} < |x|$, so $x > \frac{1}{N}$ or $x < \frac{-1}{N}$. In either case, $x \notin \left(\frac{-1}{N}, \frac{1}{N}\right)$, so $x \notin \bigcap_{n=1}^{\infty} \left(\frac{-1}{n}, \frac{1}{n}\right)$. Therefore, $\bigcap_{n=1}^{\infty} \left(\frac{-1}{n}, \frac{1}{n}\right) = \{0\}$. □

PROBLEM 32 (**) The *Fibonacci sequence* is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}$$

for $n \geq 2$. Prove the following identities for all $n \geq 0$:

$$F_{2n} = F_n(F_{n+1} - F_{n-1}) \quad F_{2n+1} = F_{n+1}^2 + F_n^2$$

Hint: Use strong induction on both identities simultaneously.

Proof. We proceed by (strong) induction on n, proving both identities simultaneously.
Base cases. For $n = 0$:

$$F_0 = 0, \qquad F_0(F_1 + F_{-1}) = 0, \qquad F_{2\cdot 0+1} = F_1 = 1 = F_1^2 + F_0^2.$$

For $n = 1$:

$$F_2 = 1 = F_1(F_2 + F_0) = 1 \cdot (1 + 0), \qquad F_3 = 2 = F_2^2 + F_1^2 = 1^2 + 1^2.$$

Inductive hypothesis. Suppose for some $n \geq 1$ that for all k with $0 \leq k \leq n$ we have

$$F_{2k} = F_k(F_{k+1} + F_{k-1}), \qquad F_{2k+1} = F_{k+1}^2 + F_k^2.$$

Inductive step. We prove the two identities for $k = n + 1$. (1) *Even index:*

$$F_{2n+2} = F_{2n+1} + F_{2n}.$$

Apply the inductive hypothesis to F_{2n+1} and F_{2n}:

$$F_{2n+2} = (F_{n+1}^2 + F_n^2) + F_n(F_{n+1} + F_{n-1}).$$

Rearrange/group terms:

$$\begin{aligned} F_{2n+2} &= F_{n+1}^2 + F_n F_{n+1} + F_n^2 + F_n F_{n-1} \\ &= F_{n+1}(F_{n+1} + F_n) + F_n(F_n + F_{n-1}). \end{aligned}$$

Using the Fibonacci recurrence $F_{n+2} = F_{n+1} + F_n$ and $F_{n+1} = F_n + F_{n-1}$,

$$F_{2n+2} = F_{n+1}F_{n+2} + F_n F_{n+1} = F_{n+1}(F_{n+2} + F_n).$$

Thus the even identity holds for $n + 1$. (2) *Odd index:*

$$F_{2n+3} = F_{2n+2} + F_{2n+1}.$$

Substitute the result just proved for F_{2n+2} and the inductive hypothesis for F_{2n+1}:

$$F_{2n+3} = F_{n+1}(F_{n+2} + F_n) + (F_{n+1}^2 + F_n^2).$$

Use $F_{n+2} = F_{n+1} + F_n$ to rewrite $F_{n+1}F_{n+2} = F_{n+1}^2 + F_{n+1}F_n$. Then

$$\begin{aligned} F_{2n+3} &= (F_{n+1}^2 + F_{n+1}F_n) + F_{n+1}F_n + F_{n+1}^2 + F_n^2 \\ &= 2F_{n+1}^2 + 2F_{n+1}F_n + F_n^2. \end{aligned}$$

On the other hand

$$F_{n+2}^2 + F_{n+1}^2 = (F_{n+1} + F_n)^2 + F_{n+1}^2 = 2F_{n+1}^2 + 2F_{n+1}F_n + F_n^2.$$

Hence

$$F_{2n+3} = F_{n+2}^2 + F_{n+1}^2.$$

so the odd identity holds for $n + 1$. Having proved both identities for $n + 1$, the inductive step is complete. By (strong) induction both identities hold for all $n \geq 0$. $\square$

PROBLEM 33 (**) The *Fibonacci sequence* is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}$$

for $n \geq 2$. Show that every positive integer can be written as the sum of distinct Fibonacci numbers. Distinct means the sum cannot have repeated values; For example, $2 = 1 + 1$ is not a valid sum even though $F_1 = 1$ and $F_2 = 1$. But $2 = 2 = F_3$ is.

Proof. We proceed by strong induction. The base cases $1 = F_1$ and $2 = F_3$ hold. Suppose for induction that all positive integers less than $n > 2$ can be written as the sum of distinct Fibonacci numbers.

We show this for n. Let F_k be the largest Fibonacci number with $F_k \leq n$. Then $F_{k+1} > n \geq F_k$. Letting $r := n - F_k$,

$$0 \leq r < F_{k+1} - F_k = F_{k-1} < n.$$

If $r = 0$, $n = F_k$ and we are done. Else, $r < n$ so by the inductive hypothesis, r can be expressed as the sum of distinct Fibonacci numbers, none of which can be F_k as $r < F_{k-1}$. Thus $n = F_k + r$ shows n can be written as the sum of distinct Fibonacci numbers.

Thus by induction the statement holds for $n \in \mathbb{N} \setminus \{0\}$. $\square$

PROBLEM 34 (**) (From the 2002 Putnam, A2). Given any five points on a sphere, show that some four of them lie in a closed hemisphere.

Proof. If none of the points are distinct, then we are done. Otherwise, choose two distinct points and create a loop or ring around the sphere that goes through both points. This splits the circle into two hemispheres. One hemisphere must have 2 of the remaining 3 points. Combined with the two points on the hemisphere's ring border, this closed hemisphere contains 4 points. $\square$

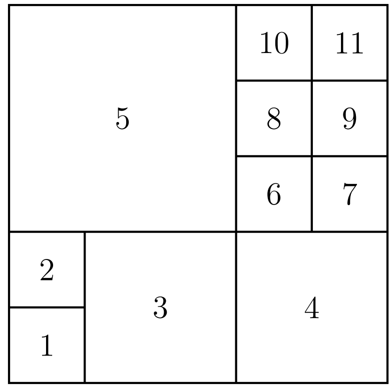
PROBLEM 35 (**) Given 51 distinct positive integers strictly less than 100, prove two of them sum to 99.

Proof. For $k = 1, 2, \ldots, 49$, define

$$P_k := \{k, 99 - k\}$$

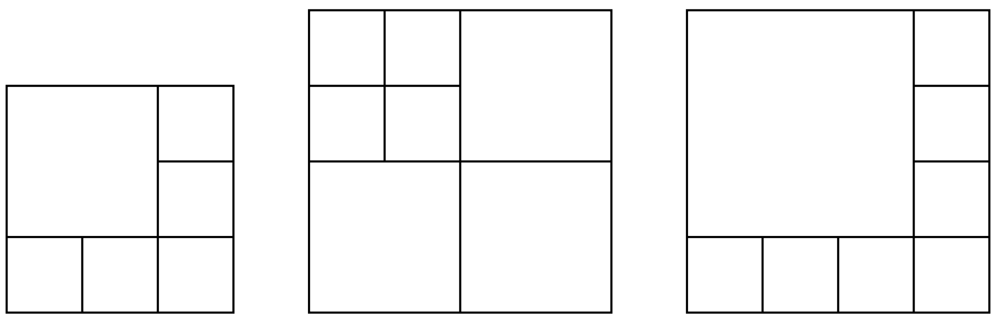
and let $P_{50} := \{99\}$. The sets $P_1, \ldots, P_{50}$ are obviously pairwise disjoint, and thus partition the set $S := \{1, 2, \ldots, 99\}$.
Suppose $C \subsetneq S$ with $|C| = 51$. At least one of the partitions $P_1, \ldots, P_{49}$ contain two elements of C, which sum to 99. Thus any 51 distinct positive integers less than 100 contain two that sum to 99. $\square$

PROBLEM 36 (**) (From this NYT Puzzle) A positive integer n is called *squareable* if we can build a square out of n smaller squares, not necessarily of the same size. For example, 11 is squareable:



Prove that all integers greater than 5 are squareable. Hint: Prove by induction with three base cases.

Proof. We show that $6, 7, 8$ are squareable:



Now suppose $n > 8$ and assume by strong induction that all integers $6 \leq k < n$ are squareable. We consider three cases:
Case 1: If $n \equiv 0 \pmod 3$, then $n = 3k$ for some integer $k \geq 3$. Take any squared tiling of $n - 3$ squares and subdivide one square into 4 equal squares.
Case 2: If $n \equiv 1 \pmod 3$, then $n = 3k + 1$ for some integer $k \geq 2$. Take any squared tiling of $n - 3$ squares and subdivide one square into 4 equal squares.
Case 3: If $n \equiv 2 \pmod 3$, then $n = 3k + 2$ for some integer $k \geq 2$. Take any squared tiling of $n - 3$ squares and subdivide one square into 4 equal squares.
In all cases, n is squareable. By induction, all integers $n \geq 6$ are squareable. □

Problem 37 (**) At a party, prove that the number of people who shook hands an odd number of times is even.

Proof. Let E be the set of people who shook hands an even number of times, and O be the set of people who shook hands an odd number of times. Let H be the total number of handshakes.
Each handshake involves exactly 2 people, so if we sum up all the handshakes each person made, we count each handshake twice:

$$\sum_{p \in E} (\text{handshakes by } p) + \sum_{p \in O} (\text{handshakes by } p) = 2H.$$

The left side has the form (even) + (sum of $|O|$ odd numbers). For the total to be even, the sum of $|O|$ odd numbers must be even. But a sum of odd numbers is even if and only if there are an even number of them. Therefore $|O|$ is even. $\square$

PROBLEM 38 (**) There exist two positive irrational numbers s and t such that s^t is rational. Hint: Take $s = t = \sqrt{2}$. Prove by cases.

PROBLEM 39 (**) For any positive integer d, there exists a number composed only of the digits 7 and 0 such that d divides this number.

Proof. Consider the $d + 1$ numbers:

$$7, 77, 777, \ldots, \underbrace{777\ldots7}_{d+1 \text{ sevens}}$$

Denote by a_n the number with n 7s. At least two must have the same remainder when divided by d. This is because there are only d possible remainders $0, 1, \ldots, d-1$ but there are $d+1$ numbers. Let a_i and a_j with $i > j$ have the same remainder modulo d. Then:

$$d \mid (a_i - a_j)$$

Furthermore,

$$\begin{aligned} a_i - a_j &= \underbrace{777\ldots7}_{i \text{ sevens}} - \underbrace{777\ldots7}_{j \text{ sevens}} \\ &= \underbrace{777\ldots7}_{j \text{ sevens}}\underbrace{000\ldots0}_{i-j \text{ zeros}} \end{aligned}$$

This is a number composed only of 7s and 0s, and d divides it. $\square$

PROBLEM 40 (**) What is the minimum number of breaks to break up a chocolate bar made of $m \times n$ little pieces into those very pieces?

Proof. Every time you break a chocolate bar, you increase the number of pieces by 1. Then you need $mn - 1$ breaks no matter what. $\square$